

## Assignment No - 01

classmate

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Subject :- Operating System

### Types of OS

#### 1) Batch OS

Definition:- Executes jobs in batches without user interaction.

##### Advantages

- High CPU utilization
- Can process many jobs automatically
- Suitable for repetitive

##### Disadvantages

- No direct user interaction
- Long waiting time
- Difficult to debug error.

#### 2) Multiprogramming OS.

Defn:- Keeps multiple programs in memory and executes them one by one to keep the CPU busy.

##### Advantages

- Better CPU utilization
- Increases system efficiency
- Reduces CPU idle time

##### Disadvantages

- Complex memory management
- Difficult to schedule
- Requires more memory.



uses:- office computers, software development data processing system.

### 3) Multitasking OS

defn:- Allows a user to run multiple application at the same time.

#### Advantages

- Improves productivity
- Save time
- Better user experience

#### Disadvantages

- Requires more RAM
- System may slow down many program.

### 4) Multiprocessing OS

defn:- Uses two or more processors to execute task simultaneously.

#### Advantages

- Fast processing
- Higher reliability
- Better performance

#### Disadvantages

- Expensive hardware
- complex system design
- Higher power consumption

### 5) Real time OS (RTOS)

defn:- process task within a fixed time limit.



### Advantages

- very Fast response
- Highly reliable
- suitable For critical system

### Disadvantages

- Expensive develop
- Limited Flexibility
- complex programming

### 6) Distributed OS.

Defn:- manages multiple connected computer as a single system.

### Advantages

- Resource sharing
- High performance
- Better reliability

### Disadvantages

- complex maintenance
- network dependency
- security challenges.

uses:- cloud computing, Banking network university and research network.

### 7) Network OS (NOS)

Defn:- manages computers connected over a network.

Advantages → i) centralized management  
ii) File, printer sharing

Disadvantages → i) server dependency  
ii) maintenance cost.



## Linux system call

### definition of system call

A system call is a mechanism through which a user program requests services from the OS. It provides an interface between user application and the OS for performing operations.

### Types of system calls.

#### 1) process control system calls

These system calls are used to create, execute, terminate and manage process.

eg] "fork()", "execve()", "exit()", "wait()"

#### 2) File management system call

These system calls are used to create, open, read, write, close and manipulate files.

eg) "open()", "read()", "write()", "close()", "lseek()", "stat()".

#### 3) device management system calls

These system calls allow communication with hardware devices such as disks, printers, keyboards, network device.



4) memory management system calls

These system calls manages memory allocation deallocation and mapping between user process and memory.

ex) "mmap()", "brk()", "munmap()"

5) Information maintenance system call

These system calls obtain or modify system and process information.

ex) "getpid()", "nice()", "getpriority()",  
"setpriority()", "brabmot2 - piling"

6) communication system calls

These system calls enable communication between processes using pipes, sockets, shared memory and message queues.

ex] "pipe()", "socket()", "send()", "

definitions of Given system calls

1) "Fork ()"

"Fork()" → create a new child process by duplicating the calling (parent) process.



2) "fork()"

→ create a child process that shares the parents address space unit it calls "execve()" or Exits.

3) "clone()"

→ create a new process of thread with shared resources depending on specified flags

4) "execve()"

→ replace the current process image with a new program.

5) "exit()"

→ Terminates a process after performing standard cleanup operation.

6) "exits()"

→ Immediately terminates a process without performing standard library cleanup.

7) "wait()"

→ suspends the parent process until one of its child process terminates

8) "waitid()"

→ wait for state change in child process and provides detailed status information.



9) "waitpid ()"

→ Wait For a specific child process to terminated

10) "kill ()"

→ send a signal to a process or process group.

11) "getpid ()"

→ Return the process ID (PID) of the calling process

12) "getppid ()"

→ Returns the parents process ID (PPID) of the calling process.

13) "nice ()"

→ changes the scheduling priority of program.

14) "setpriority ()"

→ sets the scheduling priority of a process process group or users.

15) "getpriority ()"

→ Retrieves the scheduling priority of process

16) "open ()"

→ opens an existing File or creates one if specified with appropriate flags.



17) "create ()"

→ creates a new File or truncates an existing File for writing.

18) "close ()"

→ close an open File descriptor

19) "read ()"

→ Reads data from a File for device into a buffer.

20) "write ()"

→ writes data from a specified the File offset without changing the current File

21) "pread ()"

→ writes data to a specified File offset without changing the current File pointer

22) "pwrite ()"

→ changes the current position of File using its.

23) "seek ()"

→ Retrieves information about File using its pathname.

24) "stat ()"

→ Retrieves information about a File using File descriptor.



25) "Fstat ()"

→ Retrieves information about a symbolic link without following it

26) "lstat ()"

→ checks whether a file exists and whether the process has permission to access it

27) "access ()"

→ deletes files for file system.

28) "unlink ()"

→ changes the name or location of a file or directory

29) "rename ()"

→ create a new directory

30) "mkdir ()"

→ removes an empty directory

31) "rmdir ()"

→ duplicates a file description to a space description number

32) "dup ()"

→ duplicates a file description to a space description number.



33) "dup 2 ()"

→ performs various operating of File description such as duplication locking and changing flags

34) "fcntl ()"

→ It is a system call used in Linux to perform operation on File.

35) "fsync ()"

→ writes all modified File data and metadata to permanent storage.

36) "fdatasync ()"

→ writes modified File data to disk but may delay writing unnecessary metadata

37) "truncate ()"

→ changes the size of a File using the pathname

38) "ftruncate ()"

→ changes the size of a File using its File descriptor.